Using the results of [257, Lemma 3.4], we can deduce that for any $\delta > 0$, there exists a sufficiently small $\varepsilon > 0$ such that if a realization of the Wiener process $\mathbf{W}$ remains close to our solution $\boldsymbol{R}$

$$\sup_{0 \le s \le t} \|\mathbf{W}(s) - \boldsymbol{R}(s)\| \le \varepsilon, \tag{6.50}$$

then $\mathbf{X}(t) \in \mathcal{B}_\delta(\boldsymbol{y})$. That is, if the realized Wiener process stays within a tube of radius ε centered on our derived solution $\boldsymbol{R}$ (for some sufficiently small ε), then our solution at time t is at most δ from the point $\boldsymbol{y}$.

The event (6.50) is shown in [349] to have positive probability for any $\varepsilon > 0$, and hence the required property is shown for any open set.

As Lemma 6.1 holds for all open subsets C, Assumption 1(i) holds as a corollary. Thus we have demonstrated that all of the necessary assumptions hold for Theorem 6.2. Hence Langevin dynamics is geometrically ergodic with respect to the canonical distribution

$$\rho_\beta(\boldsymbol{q},\boldsymbol{p}) = \exp(-\beta H(\boldsymbol{q},\boldsymbol{p})),$$

where it is easily checked that

$$\mathcal{L}_{\mathrm{LD}}^\dagger \rho_\beta = 0.$$

Hence averages along solutions to (6.47) will converge to the spatial average with respect to ρ_β. Thus, Langevin dynamics is a useful tool for sampling the canonical distribution, as we may sample ρ_β by computing solutions to the SDE. This remains non-trivial, as the problem usually exists in a very high dimensional space (we may consider millions of particles) with complicated non-linear potential energy function $U(\boldsymbol{q})$. Many algorithms have been proposed for computing solutions to Langevin dynamics [46, 55, 62, 265, 311]. We shall compare the sampling accuracy of some of these schemes, and address strategies for the development of discretization methods for Langevin dynamics in the following chapter.

We may also investigate the properties of the operator $\mathcal{L}_{\mathrm{LD}}$, just as we did for Brownian dynamics. However for Langevin dynamics, the lack of coupling between the noise process and all phase variables makes it relatively difficult to prove ergodicity. We denote by $\mathcal{H}^1(\mu)$ the space of functions $f(\boldsymbol{q},\boldsymbol{p})$ defined on $\mathcal{D} = \mathcal{T}^{\mathrm{N}_\mathrm{c}} \times \mathbb{R}^{\mathrm{N}_\mathrm{c}}$ with bounded norm

$$\|f\|^2_{\mathcal{H}^1(\mu)} \stackrel{\mathrm{def}}{=} \|f\|^2_{L^2(\mu)} + \|\nabla_{\boldsymbol{q}} f\|^2_{L^2(\mu)} + \|\nabla_{\boldsymbol{p}} f\|^2_{L^2(\mu)},$$

where

$$\|f\|^2_{L^2(\mu)} = \int_{\mathcal{D}} f^2 \mu(\mathrm{d}\omega),$$